Hybrid LSMC-PDE for a Generalized Gatheral Double Mean-Reverting Model

Mara Kalicanin Dimitrov[1] Ying Ni[2]

[1]Department of Business and Mathematics, Mälardalen University, SE 721 23 Västerås, Sweden
[2]Department of Business and Mathematics, Mälardalen University, SE 721 23 Västerås, Sweden

Corresponding author: Ying Ni, ying.ni@mdu.se
ORCID: Mara Kalicanin Dimitrov 0000-0000-0000-0000; Ying Ni 0000-0002-0835-7536

Abstract

To be changed
We study Bermudan-style optimal stopping problems under a generalized version of Gatheral's double mean-reverting stochastic volatility model. Our first contribution is a well-posedness analysis for the two-factor variance system: we identify parameter regimes (in particular $\delta_1, \delta_2 \geq 1/2$ under natural nonnegativity assumptions) ensuring existence of a nonnegative strong solution, which is essential for a stochastic-volatility interpretation. Our second contribution connects this model to the Mixed Least-Squares Monte Carlo-partial differential equation (LSMC-PDE) given in Farahany et al. [10] for Bermudan options. We show that the generalized double mean-reverting model satisfies the structural hypotheses needed to (i) represent conditional continuation values via lower-dimensional (random-coefficient) parabolic PDEs and (ii) obtain almost-sure convergence of the regression coefficients under standard truncation/separability conditions.

Keywords: Bermudan options, Hybrid LSMC-PDE, Gatheral Double Mean-Reverting Model

1 Introduction

The challenge of accurately pricing Bermudan options under stochastic volatility has been an important topic in financial mathematics. Bermudan options give the holder the right to exercise only at a finite set of pre-specified dates before maturity. This makes them practically relevant in structured products and also useful as a discrete approximation of American options. Under a risk-neutral measure, their value is obtained from a backward dynamic programming recursion. At each exercise date, the holder compares the immediate exercise payoff with the continuation value, which is the discounted conditional expectation of the future option value.

For low-dimensional models, deterministic numerical methods such as finite-difference, tree-based, and partial differential equation (PDE) methods can be highly accurate. The Cox–Ross–Rubinstein binomial model provides a classical discrete-time approach [6], while the Black–Scholes and Merton frameworks form the foundation of continuous-time option pricing under diffusion models [1, 20]. For Markovian models, the Feynman–Kac representation connects conditional expectations to parabolic PDEs [16]. However, when early exercise is present, the pricing problem becomes a free-boundary or variational inequality problem. This increases the numerical difficulty, especially when stochastic volatility factors are added to the state space.

Monte Carlo simulation provides a flexible alternative, particularly for high-dimensional models and path-dependent contracts [13]. For Bermudan and American options, the main difficulty is the approximation of continuation values. The least-squares Monte Carlo method of Longstaff and Schwartz [19] and the regression-based method of Tsitsiklis and Van Roy [23] address this problem by approximating conditional expectations through regression on simulated paths. These methods are simple and broadly applicable, but their performance may deteriorate in stochastic-volatility models, where the continuation surface depends on both the asset and volatility factors.

This study considers Bermudan option pricing under a generalized version of Gatheral's double mean-reverting stochastic volatility model. In the original double mean-reverting structure, the instantaneous variance mean-reverts to a stochastic variance level, and this stochastic level mean-reverts to a long-run level. This construction is motivated by the need to capture richer volatility dynamics than those available in one-factor stochastic-volatility models. In this paper, the square-root diffusion coefficients are replaced by constant elasticity of variance (CEV)-type power functions with exponents δ_1 and δ_2. The resulting generalized double mean-reverting model includes the classical square-root specification as a special case and allows more flexible volatility-of-volatility behavior.

This generalization introduces a mathematical issue near the boundary at zero. The CEV-type diffusion coefficients are not globally Lipschitz when the elasticity parameters are below one. Since the variance factors must remain nonnegative to have a valid stochastic-volatility interpretation, the first part of this paper establishes strong well-posedness and nonnegativity of the variance system. We show that when $\delta_1, \delta_2 \in [1/2, 1]$, the generalized double mean-reverting system admits a unique strong solution with nonnegative variance factors. This exponent range is natural because it is connected to the classical Yamada–Watanabe condition for pathwise uniqueness of one-dimensional SDEs with Hölder diffusion coefficients [24].

The second part of the paper develops a hybrid LSMC–PDE pricing method for the generalized model. The key structural feature is one-way coupling: the two variance factors evolve independently of the asset price. Therefore, volatility paths can be simulated first. Conditional on a simulated volatility path, the log-price equation can be reduced, after an orthogonalization of the Brownian drivers, to a one-dimensional diffusion with deterministic time-dependent coefficients. The conditional expectation required in each Bermudan step is then represented by a one-dimensional backward parabolic PDE. Solving this PDE along each volatility path produces a smooth pre-surface in the asset direction. A least-squares regression across the volatility states is then used to complete the continuation-value surface.

The hybrid construction follows the mixed LSMC–PDE framework of Farahany et al. [10]. Their method combines the variance-reduction advantages of conditional PDE pricing with the flexibility of regression Monte Carlo. The present paper verifies that the generalized double mean-reverting model satisfies the structural requirements of that framework. In particular, we prove the required one-way coupling and separability properties, derive the conditional PDE representation, and establish almost-sure convergence of the hybrid regression coefficients under the standard truncation assumptions.

The numerical part of the paper compares the Hybrid LSMC–PDE method with the plain LSMC method for Bermudan put options under a calibrated generalized double mean-reverting specification. The experiments examine both fixed-path step sweeps and fixed-step path sweeps. The benchmark prices are obtained from a high-resolution LSMC computation. The results show that the Hybrid LSMC–PDE method usually produces direct prices closer to the benchmark and with smaller standard errors and narrower confidence intervals. The improvement is most visible at intermediate step levels and moderate path budgets, while the few observed reversals occur mainly in the deepest out-of-the-money case and at small sample sizes.

The article is structured as follows. Section 2 reviews the related literature on Bermudan option pricing, stochastic-volatility models, mixed MC–PDE methods, and well-posedness of CEV-type variance processes. Section 3 introduces the generalized double mean-reverting model and the Bermudan stopping problem. Section 4 proves strong well-posedness and nonnegativity of the variance factors. Section 5 derives the conditional PDE representation used in the hybrid method. Section 6 verifies the almost-sure convergence of the hybrid regression coefficients. Section 7 describes the numerical Hybrid LSMC–PDE methodology. Section 8 presents the numerical experiments, and Section 9 concludes the paper.

2 Literature review

Bermudan and American option pricing can be formulated as optimal stopping problems under a risk-neutral measure. The value is obtained by comparing immediate exercise with the continuation value at each possible exercise date. In low-dimensional settings, PDE and finite-difference methods are often effective, since the early-exercise feature can be represented through free-boundary problems or variational inequalities. Classical contributions to American option pricing and variational inequality formulations include [21, 15]. Tree-based methods also provide an important discrete-time approach, with the binomial model of Cox, Ross, and Rubinstein serving as a standard reference [6].

Simulation-based methods are widely used when the model dimension is high or when the payoff structure is difficult to handle with deterministic grids. The least-squares Monte Carlo method introduced by Longstaff and Schwartz [19] is one of the most commonly used approaches for pricing American and Bermudan options by simulation. The related regression-based dynamic programming method of Tsitsiklis and Van Roy [23] provides another important formulation. A rigorous convergence analysis of least-squares regression methods for American option pricing was developed by Clément, Lamberton, and Protter [4]. These methods approximate continuation values from simulated paths and are therefore flexible, but their accuracy depends strongly on the quality of the regression approximation.

In stochastic-volatility models, the regression problem becomes more challenging because the state includes both the asset and volatility factors. The continuation surface may vary sharply with the volatility variables, and the exercise boundary can be difficult to estimate accurately by global regression. This motivates hybrid approaches that combine simulation and deterministic PDE methods. Mixed MC–PDE methods simulate the volatility factors and then solve conditional PDEs for the remaining asset-price uncertainty. Earlier developments of this idea include the mixed PDE/Monte Carlo approaches of Loeper and Pironneau [18] and Lipp, Loeper, and Pironneau [17]. Cozma and Reisinger [8] applied a related mixed MC–PDE variance-reduction method in a foreign-exchange setting with stochastic volatility and stochastic interest rates.

For Bermudan options, Farahany, Jackson, and Jaimungal [10] developed a hybrid LSMC–PDE method under one-way coupled stochastic-volatility models. In their framework, volatility paths are simulated independently of the asset process. Conditional on each volatility path, a lower-dimensional PDE is solved in the asset variable. This produces a smooth pre-surface, and least-squares regression across the volatility states is then used to approximate the continuation value. They also prove almost-sure convergence of the regression coefficients under truncation and separability assumptions. Farahany's thesis [9] further develops the algorithmic background, including complexity-reduction techniques and applications to multifactor stochastic-volatility models.

On the modeling side, stochastic volatility has become a standard tool for capturing features of implied-volatility surfaces that cannot be represented by the Black–Scholes model. The Heston model [14], in which the variance follows a Cox–Ingersoll–Ross square-root process [7], is one of the most widely used stochastic-volatility models. However, empirical studies show that single-factor stochastic-volatility models may be insufficient to capture the shape and term structure of the equity index option smirk. Christoffersen, Heston, and Jacobs [3] show that multifactor stochastic-volatility models can improve the fit to index option data.

Gatheral's volatility-surface framework motivates richer volatility-factor specifications [11, 12]. In particular, the double mean-reverting model describes a variance process that mean-reverts to a stochastic level, while that stochastic level itself mean-reverts to a long-run variance. This type of structure is useful for representing both short-run and long-run volatility behavior. The generalized model considered in this paper replaces the square-root volatility-of-volatility terms by CEV-type powers. This connects the model to the classical CEV specification [5] and to the broader CKLS-type family of mean-reverting diffusion models used in interest-rate modeling [2].

The use of CEV-type diffusion coefficients raises well-posedness questions near zero. Since the diffusion coefficient is only Hölder continuous, standard global Lipschitz conditions do not apply. The Yamada–Watanabe pathwise uniqueness criterion [24] is therefore central for identifying exponent ranges in which strong solutions remain well defined. For power-type diffusion coefficients, the threshold $\delta \geq 1/2$ is a natural condition for pathwise uniqueness. Recent work has also considered reflected versions of Gatheral-type double stochastic-volatility models to prevent undesirable boundary behavior [22]. The present paper complements this literature by proving strong well-posedness and nonnegativity for the generalized double mean-reverting model in the exponent regime $\delta_1, \delta_2 \in [1/2, 1]$, and by showing that this model fits into the hybrid LSMC–PDE convergence framework for Bermudan option pricing.

3 Model and Bermudan stopping problem

3.1 Stochastic basis and generalized gDMR dynamics

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}, \mathbb{Q})$ be a filtered probability space satisfying the usual conditions and carrying a 3-dimensional $(\mathcal{F}_t)$-Brownian motion $W = (W^{(1)}, W^{(2)}, W^{(3)})$ with constant correlation matrix $\rho = (\rho_{ij})_{1\leq i,j\leq 3}$, assumed symmetric positive semidefinite with $\rho_{ii} = 1$. Fix parameters $r \geq 0$ and $\kappa_1, \kappa_2, \theta, \xi_1, \xi_2 > 0$ and initial conditions $(S_0, v_0, v_0') \in (0,\infty) \times [0,\infty) \times [0,\infty)$.

We consider the generalized Gatheral double mean-reverting (gDMR) system under $\mathbb{Q}$:

$$\begin{aligned} \mathrm{d}S_t &= r\, S_t\, \mathrm{d}t + S_t \sqrt{v_t}\, \mathrm{d}W_t^{(1)}, \\ \mathrm{d}v_t &= \kappa_1\, (v_t' - v_t)\, \mathrm{d}t + \xi_1 v_t^{\delta_1}\, \mathrm{d}W_t^{(2)}, \\ \mathrm{d}v_t' &= \kappa_2\, (\theta - v_t')\, \mathrm{d}t + \xi_2 (v_t')^{\delta_2}\, \mathrm{d}W_t^{(3)}, \end{aligned} \tag{1}$$

with elasticity parameters $\delta_1, \delta_2 > 0$.

Remark 3.1 (One-way coupling). The variance subsystem (v, v') is autonomous (it does not depend on S). This one-way coupling is the structural feature exploited by mixed MC–PDE methods: volatility paths can be sampled independently of the asset, after which asset expectations are computed conditionally [10].

4 Well-posedness

4.1 Exponent regime and strong well-posedness

Assumption 4.1 (CEV exponents)**.** We impose $\delta_1, \delta_2 \in [1/2, 1]$.

Theorem 4.2 (Strong existence/uniqueness and positivity)**.** *Assume Assumption 4.1. Then for any $T > 0$ the system* (1) *admits a unique strong solution $(S_t, v_t, v'_t)_{t\in[0,T]}$ with continuous paths such that*

$$S_t > 0, \qquad v_t \geq 0, \qquad v'_t \geq 0, \qquad \forall\, t \in [0,T] \quad \mathbb{Q}\text{-}a.s.$$

Proof. **Well-posedness and nonnegativity of v'.** Define globally measurable coefficients on $\mathbb{R}$ by

$$b_2(x) := \kappa_2(\theta - x), \qquad \sigma_2(x) := \xi_2(x^+)^{\delta_2}, \qquad x^+ := \max(x, 0).$$

Then b_2 is globally Lipschitz with linear growth, and σ_2 is Hölder of order δ_2. For $\delta_2 \geq 1/2$, the Yamada–Watanabe modulus condition holds, implying pathwise uniqueness for the one-dimensional SDE

$$\mathrm{d}v'_t = b_2(v'_t)\mathrm{d}t + \sigma_2(v'_t)\mathrm{d}W_t^{(3)}, \qquad v'_0 \geq 0;$$

see Yamada and Watanabe [24] and Karatzas and Shreve [16, Ch. 5]. Since the coefficients are continuous with linear growth, weak existence holds, and hence a unique strong solution exists.

Well-posedness and nonnegativity of v given v'. Given the adapted continuous process v', define

$$b_1(t, x) := \kappa_1(v'_t - x), \qquad \sigma_1(x) := \xi_1(x^+)^{\delta_1}.$$

For each (t, ω) the map $x \mapsto b_1(t, x)$ is globally Lipschitz with constant κ_1, and σ_1 is Hölder of order $\delta_1 \geq 1/2$. The same Yamada–Watanabe argument (now with time-dependent but Lipschitz drift) yields a unique strong solution v to

$$\mathrm{d}v_t = b_1(t, v_t)\mathrm{d}t + \sigma_1(v_t)\mathrm{d}W_t^{(2)}.$$

Explicit solution for S and strict positivity. With v nonnegative and progressively measurable, the asset SDE is linear in S and admits the unique strong solution

$$S_t = S_0 \exp\left(\int_0^t \left(r - \tfrac{1}{2}v_s\right)\mathrm{d}s + \int_0^t \sqrt{v_s}\,\mathrm{d}W_s^{(1)}\right),$$

which is strictly positive for all $t \in [0,T]$ whenever $S_0 > 0$. □

Remark 4.3 (Why $\delta_i \geq 1/2$ is natural). The lower bound $\delta_i \geq 1/2$ is a classical threshold for pathwise uniqueness in CEV-type diffusions: for

$$\mathrm{d}X_t = b(X_t)\mathrm{d}t + \sigma(X_t^+)^{\delta}\mathrm{d}W_t,$$

the Yamada–Watanabe modulus condition holds when $\delta \geq 1/2$, while for $\delta < 1/2$ pathwise uniqueness can fail [24]. Since the hybrid MC–PDE convergence results of Farahany et al. [10] are formulated for models admitting strong unique solutions, Assumption 4.1 is a minimal structural restriction that keeps the gDMR model within that framework.

4.2 Bermudan option valuation as a discrete-time stopping problem

Let $\pi = \{0 = t_0 < \cdots < t_M = T\}$ be the exercise grid and let $(h_n)_{n=0}^M$ be measurable payoff functions with $h_n : (0,\infty) \to \mathbb{R}$. For simplicity, we assume that the payoff at time t_n depends only on S_{t_n}. The Bermudan value process $(V_n)_{n=0}^M$ satisfies the Snell recursion

$$V_M = h_M(S_{t_M}), \qquad V_n = \max\Big\{h_n(S_{t_n}),\, e^{-r(t_{n+1}-t_n)}\,\mathbb{E}^{\mathbb{Q}}\big[V_{n+1} \mid \mathcal{F}_{t_n}\big]\Big\}, \qquad n = M-1, \ldots, 0. \tag{2}$$

Since (S_t, v_t, v'_t) is Markov under $\mathbb{Q}$, the continuation value can be written as a deterministic function of the current state. Specifically, define

$$C_n(s, v, v') := e^{-r(t_{n+1}-t_n)}\,\mathbb{E}^{\mathbb{Q}}\big[V_{n+1}\big(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}\big)\,\big|\, S_{t_n} = s,\ v_{t_n} = v,\ v'_{t_n} = v'\big]. \tag{3}$$

Then the one-step continuation value appearing in (2) is given by

$$e^{-r(t_{n+1}-t_n)}\,\mathbb{E}^{\mathbb{Q}}\big[V_{n+1} \mid \mathcal{F}_{t_n}\big] = C_n\big(S_{t_n}, v_{t_n}, v'_{t_n}\big) \qquad \mathbb{Q}\text{-a.s.}$$

5 Conditional PDE representation

Since the variance factors (v, v') are *one-way coupled* (their dynamics do not depend on S), the entire volatility path over an interval $[t_n, t_{n+1}]$ is measurable with respect to the Brownian drivers that generate (v, v'). After an orthogonalization of the *asset* Brownian motion, the remaining randomness in the log-price evolution on $[t_n, t_{n+1}]$ is carried by a Brownian motion that is independent of those volatility drivers. Conditional expectations over one exercise step therefore reduce to a one-dimensional (time-inhomogeneous) diffusion problem, which is characterized by a backward parabolic PDE with *random (pathwise) coefficients* Farahany et al. [10].

5.1 Orthogonalization on a Bermudan step

Fix an interval $[t_n, t_{n+1}]$ in a Bermudan exercise grid and define

$$\mathcal{G}_n := \sigma\big((W_s^{(2)}, W_s^{(3)}) \ : \ s \in [t_n, t_{n+1}]\big).$$

Since the variance subsystem is one-way coupled and driven by $(W^{(2)}, W^{(3)})$, the path $(v_s, v'_s)_{s\in[t_n,t_{n+1}]}$ is measurable with respect to $\mathcal{F}_{t_n} \vee \mathcal{G}_n$.

Assume $|\rho_{23}| < 1$ and define

$$\Sigma_{23} := \begin{bmatrix} 1 & \rho_{23} \\ \rho_{23} & 1 \end{bmatrix}, \qquad \beta := \Sigma_{23}^{-1} \begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} = \begin{bmatrix} \beta_2 \\ \beta_3 \end{bmatrix}.$$

Equivalently,

$$\beta_2 = \frac{\rho_{12} - \rho_{13}\rho_{23}}{1 - \rho_{23}^2}, \qquad \beta_3 = \frac{\rho_{13} - \rho_{12}\rho_{23}}{1 - \rho_{23}^2}. \tag{4}$$

The conditional variance of $W^{(1)}$ after projection onto $\operatorname{span}\{W^{(2)}, W^{(3)}\}$ is

$$\sigma_\perp^2 := 1 - \begin{bmatrix} \rho_{12} & \rho_{13} \end{bmatrix} \Sigma_{23}^{-1} \begin{bmatrix} \rho_{12} \\ \rho_{13} \end{bmatrix} = \frac{1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23}}{1 - \rho_{23}^2} \geq 0. \tag{5}$$

Hence there exists a Brownian motion $B^\perp$, independent of $(W^{(2)}, W^{(3)})$, such that on $[t_n, t_{n+1}]$,

$$\mathrm{d}W_t^{(1)} = \beta_2 \, \mathrm{d}W_t^{(2)} + \beta_3 \, \mathrm{d}W_t^{(3)} + \sigma_\perp \, \mathrm{d}B_t^\perp. \tag{6}$$

Let $X_t := \log S_t$. Under $\mathbb{Q}$, we have

$$\mathrm{d}X_t = \Big(r - \tfrac{1}{2} v_t\Big)\mathrm{d}t + \sqrt{v_t} \, \mathrm{d}W_t^{(1)}. \tag{7}$$

Substituting (6) into (7) yields, on $[t_n, t_{n+1}]$,

$$\mathrm{d}X_t = \Big(r - \tfrac{1}{2} v_t\Big)\mathrm{d}t + \sqrt{v_t}\Big(\beta_2 \, \mathrm{d}W_t^{(2)} + \beta_3 \, \mathrm{d}W_t^{(3)}\Big) + \sigma_\perp \sqrt{v_t} \, \mathrm{d}B_t^\perp. \tag{8}$$

Define the pathwise correlated shift and the orthogonal log component

$$Z_t^{\text{shift}} := \int_{t_n}^{t} \sqrt{v_s} \, \big(\beta_2 \, \mathrm{d}W_s^{(2)} + \beta_3 \, \mathrm{d}W_s^{(3)}\big), \qquad Y_t := X_t - Z_t^{\text{shift}}. \tag{9}$$

Then $X_t = Y_t + Z_t^{\text{shift}}$ and, on $[t_n, t_{n+1}]$,

$$\mathrm{d}Y_t = \Big(r - \tfrac{1}{2} v_t\Big)\mathrm{d}t + \sigma_\perp \sqrt{v_t} \, \mathrm{d}B_t^\perp. \tag{10}$$

In the degenerate case $\sigma_\perp = 0$, the stock Brownian driver lies in the span of the volatility Brownian drivers and, conditional on the volatility path, the propagation of X becomes deterministic.

5.2 Conditional expectation as a random-coefficient PDE

Lemma 5.1. *Assume Assumption 4.1 and let* $G : \mathbb{R} \times [0, \infty)^2 \to \mathbb{R}$ *be bounded and Borel measurable. Fix* $n \in \{0, \ldots, M-1\}$ *and work on the interval* $[t_n, t_{n+1}]$*. Let*

$$\mathcal{G}_n := \sigma\big((W_s^{(2)}, W_s^{(3)}) : s \in [t_n, t_{n+1}]\big),$$

and define

$$Z_n := \int_{t_n}^{t_{n+1}} \sqrt{v_s}\,(\beta_2 \, \mathrm{d}W_s^{(2)} + \beta_3 \, \mathrm{d}W_s^{(3)}),$$

where β_2, β_3 *are given by* (4) *and* $\sigma_\perp$ *by* (5)*. For* $t \in [t_n, t_{n+1}]$ *and* $y \in \mathbb{R}$ *set*

$$u(t, y) := \mathbb{E}^{\mathbb{Q}}\Big[G\big(Y_{t_{n+1}} + Z_n,\ v_{t_{n+1}},\ v'_{t_{n+1}}\big)\,\Big|\,\mathcal{G}_n,\ Y_t = y\Big],$$

where Y *is the orthogonal log-component in* (10) *with* $Y_{t_n} = \log S_{t_n}$*.*

Then, conditional on $\mathcal{G}_n$*,* u *is the bounded solution of the random-coefficient backward PDE*

$$\begin{cases} \partial_t u(t, y) + \Big(r - \frac{1}{2} v_t\Big)\partial_y u(t, y) + \frac{1}{2}\sigma_\perp^2 v_t \, \partial_{yy} u(t, y) = 0, & (t, y) \in [t_n, t_{n+1}) \times \mathbb{R}, \\ u(t_{n+1}, y) = G\big(y + Z_n,\ v_{t_{n+1}},\ v'_{t_{n+1}}\big), & y \in \mathbb{R}. \end{cases}$$

Proof. By (6)–(10), on $[t_n, t_{n+1}]$ we have

$$\mathrm{d}Y_t = \Big(r - \frac{1}{2} v_t\Big)\mathrm{d}t + \sigma_\perp \sqrt{v_t}\, \mathrm{d}B_t^\perp,$$

where $B^\perp$ is a Brownian motion independent of $(W^{(2)}, W^{(3)})$. Since the variance subsystem (v, v') is driven by $(W^{(2)}, W^{(3)})$, the entire path $t \mapsto v_t$ on $[t_n, t_{n+1}]$ is $\mathcal{G}_n$-measurable. Hence, conditional on $\mathcal{G}_n$, Y is a time-inhomogeneous one-dimensional diffusion with *deterministic* coefficients, and the conditional Feynman-Kac formula yields that u solves the stated backward parabolic PDE with terminal condition at t_{n+1}. □

6 Almost-sure convergence of the hybrid regression coefficients

This section adapts the almost-sure convergence theorem of Farahany et al. [10] to the generalized gDMR model (1). Our goal is to (i) restate the regression coefficients in a self-contained manner (in the present two-factor volatility setting), and (ii) verify, in detail, that the structural hypotheses required by the Farahany et al. [10] framework are satisfied.

Throughout, we work under the risk-neutral measure $\mathbb{Q}$ and the time grid $\pi = \{0 = t_0 < t_1 < \cdots < t_M = T\}$ used in the Bermudan recursion.

6.1 Regression set-up and truncation conditions

Let $d_B \in \mathbb{N}$ and let

$$\phi = (\phi_1, \ldots, \phi_{d_B}) : [0, \infty)^2 \to \mathbb{R}^{d_B}$$

be a fixed vector of basis functions in the two-factor volatility state (v, v'). Let $\mathcal{D}_v \subset [0, \infty)^2$ be a compact rectangle such that $\mathrm{supp}(\phi) \subset \mathcal{D}_v$, and let $\mathcal{S} \subset (0, \infty)$ be a compact set containing the support of every exercise function h_n (this is the standard spatial truncation used when solving the conditional PDEs).

Idealized continuation functions. Following Farahany et al. [10], define the *idealized* continuation functions by backward induction:

$$c_M(s, v, v') \equiv 0, \qquad c_n(s, v, v') := a_n(s) \cdot \phi(v, v'), \qquad n = 0, \ldots, M-1,$$

where, for $n < M$, the coefficient vector $a_n(s) \in \mathbb{R}^{d_B}$ is obtained by least-squares regression of the next-step value. Define the associated "value functions"

$$f_n(s, v, v') := \max\{h_n(s),\ c_n(s, v, v')\}, \qquad n = 0, \ldots, M-1, \qquad \text{and} \qquad f_M(s, v, v') := h_M(s).$$

For fixed (n, s), define the conditional least-squares objective

$$H_{n,s}(a) := \mathbb{E}_{\mathbb{Q}}\Big[\Big(\mathbb{E}_{\mathbb{Q}}\big[f_{n+1}(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \mid S_{t_n} = s,\ v_{t_n},\ v'_{t_n}\big] - a \cdot \phi(v_{t_n}, v'_{t_n})\Big)^2 \ \Big|\ S_{t_n} = s\Big]. \tag{11}$$

The (idealized) coefficient vector is any minimizer

$$a_n(s) \in \arg\min_{a\in\mathbb{R}^{d_B}} H_{n,s}(a).$$

Under the truncation conditions below, $H_{n,s}$ is finite and the normal equations yield the explicit representation:

$$a_n(s) = A_n^{-1}\, b_n(s), \qquad A_n := \mathbb{E}_{\mathbb{Q}}\Big[\phi(v_{t_n}, v'_{t_n})\, \phi(v_{t_n}, v'_{t_n})^\top\Big], \qquad b_n(s) := \mathbb{E}_{\mathbb{Q}}\Big[\phi(v_{t_n}, v'_{t_n})\, f_{n+1}(S_{t_{n+1}}, v_{t_{n+1}}, v'_{t_{n+1}}) \;\Big|\; S_{t_n} = s\Big]. \tag{12}$$

Estimated continuation functions (hybrid LSMC–PDE). Let $\mathbb{Q}'$ denote the inherited sampling measure used to construct i.i.d. copies of the volatility path segments. Under $\mathbb{Q}'$, let $\{(v^j_{t_k}, v'^j_{t_k})_{k=0}^M\}_{j\geq 1}$ be i.i.d. copies of the volatility skeleton, and let $[v^j, v'^j]_{t_n}^{t_{n+1}}$ denote the corresponding path segment on $[t_n, t_{n+1}]$.

Define the estimated continuation functions recursively by

$$c_M^N \equiv 0, \qquad c_n^N(s,v,v') := a_n^N(s)\cdot\phi(v,v'), \qquad f_n^N(s,v,v') := \max\{h_n(s),\, c_n^N(s,v,v')\}, \qquad f_M^N := h_M.$$

At time step n, the hybrid algorithm computes the pre-regression conditional expectations (via the conditional PDE) along each sampled volatility segment:

$$\bar{C}_n^{j,N}(s) := \mathbb{E}_{\mathbb{Q}}\Big[f_{n+1}^N(S_{t_{n+1}}, v^j_{t_{n+1}}, v'^j_{t_{n+1}}) \;\Big|\; S_{t_n} = s,\ [v^j, v'^j]_{t_n}^{t_{n+1}}\Big].$$

The regression matrix and right-hand side are then

$$A_n^N := \frac{1}{N}\sum_{j=1}^N \phi(v^j_{t_n}, v'^j_{t_n})\, \phi(v^j_{t_n}, v'^j_{t_n})^\top, \qquad b_n^N(s) := \frac{1}{N}\sum_{j=1}^N \phi(v^j_{t_n}, v'^j_{t_n})\, \bar{C}_n^{j,N}(s),$$

and the (truncated) hybrid regression coefficient is

$$a_n^N(s) := \big[A_n^N\big]_R^{-1}\, b_n^N(s), \qquad \big[A_n^N\big]_R^{-1} := \big[A_n^N\big]^{-1}\, \mathbf{1}_{\{\|[A_n^N]^{-1}\|\leq R\}}, \tag{13}$$

for a fixed truncation constant $R > 0$ chosen as in the standard truncation scheme. Using $\mathbf{1}_{\{\|[A_n^N]^{-1}\|\leq R\}}$ is ensuring uniform boundedness of the inverse.

Assumption 6.1 (Truncation conditions)**.**

1. The basis functions $\{\phi_m\}_{m=1}^{d_B}$ are bounded and supported on the compact rectangle $\mathcal{D}_v \subset [0,\infty)^2$.
2. For each n, the exercise function h_n is bounded and has compact support contained in $\mathcal{S} \subset (0,\infty)$.
3. The regression inverse is truncated as in (13) for some $R > 0$, so that $\sup_{n\leq M-1,\, N\geq 1} \|[A_n^N]_R^{-1}\| < \infty$ holds $\mathbb{Q}'$-a.s.

Remark Item 2) holds automatically on any finite PDE grid because h_n is evaluated only on the truncated domain $\mathcal{S}$ ($[s_{\min}, s_{max}]$). For common payoffs without compact support (e.g. put), one may apply a standard cutoff: truncate the payoff beyond a large threshold and extend continuously to 0 outside a slightly larger threshold. Likewise, common bases (polynomials, etc.) may be multiplied by a smooth cutoff so that $\mathrm{supp}(\phi) \subset \mathcal{D}_v$ while keeping $\mathcal{D}_v$.

6.2 Separability and finite-dimensional path statistics

The almost-sure convergence proof of Farahany et al. [10] relies on separability of the asset dynamics and on the fact that the conditional expectations used in the hybrid algorithm depend on each volatility path segment only through a *finite-dimensional* statistic.

Proposition 6.2 (Multiplicative return representation)**.** *Fix $0 \leq t < u \leq T$. The solution of* (1) *satisfies*

$$S_u = S_t\, R_{t,u}, \qquad R_{t,u} := \exp\!\left(\int_t^u \Big(r - \tfrac{1}{2}v_s\Big)\, ds + \int_t^u \sqrt{v_s}\, dW_s^{(1)}\right).$$

In particular, for fixed (t,u), the factor $R_{t,u}$ does not depend on the value of S_t.

Proof. Apply the explicit stochastic exponential representation for S from Theorem 4.2 on $[t,u]$. □

Assumption 6.3 (Separability conditions)**.** 1. $S_t > 0$ for all $t \in [0, T]$ almost surely.

2. For each exercise step n, if $S_{t_n} = s$ then $S_{t_{n+1}} = s\, R_n$, where R_n does not depend on s.

3. The exercise functions h_n are continuous with compact support in $(0, \infty)$, and the basis functions ϕ are continuous with compact support in $[0, \infty)^2$.

Comment. Assumption 6.3(1)–(2) is a model property: it holds for (1) by Theorem 4.2 and Proposition 6.2. Assumption 6.3

6.3 Main convergence theorem

Theorem 6.4 (Almost-sure coefficient convergence for the gDMR model)**.** *Assume Assumption 4.1, and Assumptions 6.1 and 6.3. Fix $n \in \{0, \dots, M-1\}$ and $s \in \mathcal{S}$. Let $a_n(s)$ be the idealized coefficient vector defined by* (11)–(12)*, and let $a_n^N(s)$ be the hybrid LSMC–PDE estimator defined by* (13) *from N i.i.d. volatility samples under the inherited sampling measure $\mathbb{Q}'$. Then*

$$\lim_{N\to\infty} \left\| a_n^N(s) - a_n(s) \right\| = 0 \qquad \mathbb{Q}'\text{-almost surely.}$$

Consequently, the estimated continuation functions $c_n^N(s, v, v') := a_n^N(s) \cdot \phi(v, v')$ converge uniformly *on the truncation set $\mathcal{D}_v$:*

$$\lim_{N\to\infty} \sup_{(v,v')\in\mathcal{D}_v} \left| c_n^N(s, v, v') - c_n(s, v, v') \right| = 0, \qquad \mathbb{Q}'\text{-almost surely.}$$

In particular, $f_n^N(s, v, v') = \max\{h_n(s), c_n^N(s, v, v')\}$ converges uniformly to $f_n(s, v, v')$ on $\mathcal{D}_v$.

Proof. We verify the hypotheses of the almost-sure convergence theorem of Farahany et al. [10] in the present two-factor volatility setting.

Pure stochastic volatility / one-way coupling. By construction, the volatility subsystem (v_t, v'_t) is autonomous and does not depend on S. Hence, volatility path segments can be sampled independently of the current asset level, and the inherited sampling space $(\Omega', \mathcal{F}', \mathbb{Q}')$ of Farahany et al. [10] applies.

Well-posedness. Under Assumption 4.1, Theorem 4.2 yields a unique strong solution of (1) with $S_t > 0$ and $v_t, v'_t \geq 0$ on $[0, T]$ almost surely. This provides the basic existence/uniqueness and positivity required by the framework.

Truncation scheme. Assumption 6.1 is exactly the truncation scheme used in [10]: bounded compactly supported payoff/basis functions make the objectives (11) finite, and the truncated inverse $[A_n^N]_R^{-1}$ makes $a_n^N(s)$ well-defined and uniformly bounded in (n, N).

Separability. Assumption 6.3(1) holds by positivity of S from Theorem 4.2. Assumption 6.3(2) holds by Proposition 6.2 with $R_n = R_{t_n,t_{n+1}}$. Assumption 6.3(3) is imposed by truncation and matches the compact-support setting required in [10].

Apply Farahany et al. Having verified one-way coupling, truncation, separability, and the finite-statistic property, we may invoke Theorem 1 of Farahany et al. [10](with volatility state $(v_{t_n}, v'_{t_n}) \in [0, \infty)^2$) to conclude that $\|a_n^N(s) - a_n(s)\| \to 0$ $\mathbb{Q}'$-almost surely.

Uniform convergence of $c_n^N(s, \cdot, \cdot)$ on $\mathcal{D}_v$ follows immediately from boundedness of ϕ on $\mathcal{D}_v$:

$$\sup_{(v,v')\in\mathcal{D}_v} |(a_n^N(s) - a_n(s)) \cdot \phi(v, v')| \leq \|a_n^N(s) - a_n(s)\| \sup_{(v,v')\in\mathcal{D}_v} \|\phi(v, v')\|.$$

Finally, uniform convergence of f_n^N follows from the 1-Lipschitz property of $x \mapsto \max\{h_n(s), x\}$. □

7 A hybrid LSMC-PDE methodology for the gDMR model

This section summarizes a computational hybrid method for Bermudan pricing under the one-way coupled gDMR model (1). The method follows the mixed LSMC-PDE philosophy of Farahany et al. Farahany et al. [10]: (i) simulate the autonomous volatility factors, (ii) compute the *one-step* conditional expectations in the Snell recursion by solving a *one-dimensional* random-coefficient PDE in the log-price variable along each sampled volatility path, producing a smooth *pre-surface* over the asset grid, and (iii) regress these pre-surface values across volatility states to obtain a completed continuation-value surface over the volatility state space.

7.1 Discretization and regression ansatz

Let $\pi = \{0 = t_0 < \cdots < t_M = T\}$ be the Bermudan exercise grid and $(h_n)_{n=0}^M$ the (bounded, compactly supported) exercise payoff functions. We fix a spatial grid of asset values

$$\mathcal{S} := \{s_1, \dots, s_{N_S}\} \subset (0, \infty), \qquad y_i := \log s_i,$$

which is used by the conditional PDE solver in log-space. We also fix a truncation rectangle $D_v \subset [0, \infty)^2$ in the volatility state space and choose d_B bounded basis functions

$$\phi = (\phi_1, \dots, \phi_{d_B}) \quad \text{supported on } D_v.$$

Following Farahany et al. Farahany et al. [10], we approximate the continuation value at time t_n in the separable form

$$c_n(s, v, v') \approx a_n(s) \cdot \phi(v, v'), \qquad a_n(s) \in \mathbb{R}^{d_B}. \tag{14}$$

In practice we compute $a_n(s)$ only on the grid $s \in \mathcal{S}$, and use interpolation in s when evaluating the approximation at off-grid asset values.

We truncate the volatility state space by fixing a compact rectangle $D_v = [0, \bar{v}] \times [0, \bar{v}'] \subset [0, \infty)^2$, where $\bar{v}, \bar{v}'$ are chosen as high empirical quantiles of $\{(v_{t_n}^j, (v_{t_n}')^j)\}$ from the volatility simulation.

7.2 Pathwise conditional PDE step and pre-surfaces

Fix a time index $n \in \{0, \dots, M-1\}$ and assume we have already constructed an approximation $\widehat{V}_{n+1}$ of the value function at t_{n+1} on $\mathcal{S} \times D_v$. For each Monte Carlo sample $j \in \{1, \dots, N\}$ we simulate a volatility path segment on $[t_n, t_{n+1}]$,

$$\left(v_t^j, (v_t')^j\right)_{t \in [t_n, t_{n+1}]},$$

using a fine internal time discretization on the interval. Along with the volatility segment we compute the pathwise correlated shift

$$Z_n^j := \int_{t_n}^{t_{n+1}} \sqrt{v_s^j}\left(\beta_2 \, dW_s^{(2),j} + \beta_3 \, dW_s^{(3),j}\right), \tag{15}$$

where β_2, β_3 are given by (4) and $\sigma_\perp$ by (5). By Lemma 5.1 (Section 5), conditional on the volatility drivers on $[t_n, t_{n+1}]$ the orthogonalized log-price component Y evolves as a one-dimensional diffusion with deterministic coefficients,

$$dY_t = \left(r - \tfrac{1}{2} v_t^j\right) dt + \sigma_\perp \sqrt{v_t^j} \, dB_t^\perp, \qquad t \in [t_n, t_{n+1}],$$

and the conditional expectation required for the continuation value may be written as the solution of a backward parabolic PDE in (t, y). Concretely, define the terminal function on log-space by

$$g_{n+1}^j(y) := \widehat{V}_{n+1}\left(e^y, \, v_{t_{n+1}}^j, \, (v_{t_{n+1}}')^j\right),$$

and let $u_n^j(t, y)$ denote the conditional expectation

$$u_n^j(t, y) := \mathbb{E}^{\mathbb{Q}}\left[g_{n+1}^j\left(Y_{t_{n+1}} + Z_n^j\right) \,\middle|\, Y_t = y, \; (v^j, (v')^j)_{t_n}^{t_{n+1}}\right].$$

Then u_n^j satisfies, on $[t_n, t_{n+1}) \times \mathbb{R}$,

$$\partial_t u + \left(r - \tfrac{1}{2} v_t^j\right) \partial_y u + \tfrac{1}{2} \sigma_\perp^2 v_t^j \, \partial_{yy} u = 0, \qquad u(t_{n+1}, y) = g_{n+1}^j(y + Z_n^j). \tag{16}$$

The *pre-surface* is obtained by evaluating the PDE solution at $t = t_n$ on the spatial grid:

$$\widehat{C}_n^j(s_i) := e^{-r \Delta t_n} \, u_n^j(t_n, y_i), \qquad i = 1, \dots, N_{\mathcal{S}}, \tag{17}$$

where $\Delta t_n = t_{n+1} - t_n$. This provides, for each volatility sample j, a smooth function of $s \in \mathcal{S}$.

Because (16) has coefficients depending only on time (through v_t^j), it is translation-invariant in y and admits a closed-form solution in Fourier space (cf. Farahany et al. [10]). Let

$$A_n^j := \int_{t_n}^{t_{n+1}} \left(r - \tfrac{1}{2} v_s^j\right) ds, \qquad B_n^j := \sigma_\perp^2 \int_{t_n}^{t_{n+1}} v_s^j \, ds.$$

Taking the Fourier transform in y reduces the PDE to an ODE in time, yielding the characteristic exponent

$$\Psi_n^j(\omega) := i\omega Z_n^j + i\omega A_n^j - \tfrac{1}{2}\omega^2 B_n^j.$$

On a uniform log-grid in y, FST computes the recursion

$$u_n^j(t_n, \cdot) \;\approx\; \mathrm{FFT}^{-1}\Big(\mathrm{FFT}[g_{n+1}^j]\ \exp(\Psi_n^j)\Big), \tag{18}$$

where the exponential is applied pointwise in Fourier space. In the present one-dimensional setting this produces all values $u_n^j(t_n, y_i)$ simultaneously, after the pathwise scalars (Z_n^j, A_n^j, B_n^j) have been computed from the volatility discretization.

7.3 Regression across volatility states and backward recursion

Having computed the pre-surface realizations $\{\widehat{C}_n^j(s_i)\}_{j=1}^N$ at each fixed $s_i \in \mathcal{S}$, we regress across the volatility states at time t_n to complete the continuation surface. Specifically, for each s_i we compute

$$a_n^N(s_i) \in \arg\min_{a \in \mathbb{R}^{d_B}} \sum_{j=1}^N \Big|\widehat{C}_n^j(s_i) - a \cdot \phi\big(v_{t_n}^j, (v'_{t_n})^j\big)\Big|^2. \tag{19}$$

The resulting completed continuation-value surface on the grid is

$$\widehat{C}_n^N(s_i, v, v') := a_n^N(s_i) \cdot \phi(v, v'),$$

and the value approximation at time t_n is given by the Bermudan recursion

$$\widehat{V}_n^N(s_i, v, v') := \max\{h_n(s_i),\ \widehat{C}_n^N(s_i, v, v')\}.$$

Starting from $\widehat{V}_M^N(s_i, v, v') := h_M(s_i)$ and iterating backward for $n = M-1, \dots, 1$ produces an approximation of the value and continuation surfaces on $\mathcal{S} \times D_v$. If we consider the Bermudan put option, then $h_n(s) = (K-s)^+$ for all exercise dates t_n.

7.4 Time-zero estimators (direct and low)

At time $t_0 = 0$ the volatility state (v_0, v'_0) is fixed, so the time-zero continuation value may be estimated by averaging the first-step pre-surface values:

$$\widehat{C}_0(s_i) := \frac{1}{N}\sum_{j=1}^N \widehat{C}_0^j(s_i), \qquad i = 1, \dots, N_{\mathcal{S}},$$

and defining the *direct* estimator

$$\widehat{V}_{d,0}(s_i) := \max\{h_0(s_i),\ \widehat{C}_0(s_i)\}.$$

To obtain a (typically) low-biased estimator, we use the estimated continuation surfaces to define a suboptimal stopping rule on the exercise grid. For an off-grid state $(S_{t_n}, v_{t_n}, v'_{t_n})$, evaluate $\widehat{C}_n^N(S_{t_n}, v_{t_n}, v'_{t_n})$ by interpolating $a_n^N(\cdot)$ in s and forming the dot product with $\phi(v_{t_n}, v'_{t_n})$. Define the stopping index

$$\tau^N := \inf\Big\{n \in \{0, \dots, M\} : h_n(S_{t_n}) \ge \widehat{C}_n^N(S_{t_n}, v_{t_n}, v'_{t_n})\Big\},$$

with the convention that $\tau^N = M$ if the set is empty. Simulating N_{low} independent sample paths of the full state process on the exercise grid and applying this policy yields the *low* estimator

$$\widehat{V}_{l,0} := \frac{1}{N_{\text{low}}}\sum_{k=1}^{N_{\text{low}}} e^{-rt_{\tau_k^N}}\ h_{\tau_k^N}\big(S_{t_{\tau_k^N}}^{(k)}\big).$$

8 Numerical Studies

This section presents the direct Bermudan pricing experiments under the calibrated gDMR specification. The plain LSMC method is compared with the Hybrid LSMC–PDE method. Two types of experiments are considered. The first keeps the direct path budget fixed and varies the number of steps. The second keeps the number of steps fixed and varies the direct path budget. Throughout this section, only direct prices are reported, together with standard errors, confidence intervals, and relative errors with respect to benchmark reference values.

8.1 Experimental setting and benchmark

Table 1 reports the calibrated model parameters and the fixed numerical setting used in all Bermudan experiments. Since $S_0 = 100$, the strikes $K = 70$, $K = 80$, and $K = 90$ are out-of-the-money, $K = 100$ is at-the-money, and $K = 110$ is in-the-money for the Bermudan put. The Hybrid LSMC–PDE settings are kept fixed throughout both study families.

Table 1: Calibrated gDMR parameters and fixed numerical setting used in the Bermudan pricing experiments.

Symbol	Value	Interpretation
Model and contract parameters		
S_0	100	initial asset level
v_0	0.114	initial instantaneous variance factor
v_0'	0.110	initial secondary variance factor
r	0	risk-free rate
T	1	maturity in years
κ_1	5.5	mean-reversion speed of v_t toward v_t'
κ_2	0.1	mean-reversion speed of v_t' toward θ
θ	0.078	long-run variance level
ξ_1	2.689	volatility scale of the first variance factor
ξ_2	0.502	volatility scale of the second variance factor
δ_1	0.94	CEV exponent of the first variance factor
δ_2	0.94	CEV exponent of the second variance factor
ρ_{12}	-0.982	correlation between the first and second Brownian drivers
ρ_{13}	-0.727	correlation between the first and third Brownian drivers
ρ_{23}	0.590	correlation between the second and third Brownian drivers
Fixed numerical setting		
N_{ex}	12	Bermudan exercise dates
K	$\{70, 80, 90, 100, 110\}$	strike set used in all experiments
N_S	301	asset-grid points in the conditional PDE solver
$(a_{\min}, a_{\max})$	$(0.30, 3.50)$	asset-range factors relative to S_0
q_v	0.999	volatility truncation quantile
$(N_{\text{ref}}, M_{\text{ref}})$	(1200, 1,200,000)	benchmark LSMC step count and path budget

Benchmark reference prices are reported separately in Table 2. The Hybrid LSMC–PDE settings are kept fixed throughout the two study families.

All reported quantities are direct estimators. The benchmark is obtained from the plain LSMC method with 1200 steps and 1,200,000 paths. For a direct estimator V and benchmark reference value V^{ref}, the direct relative error is defined by

$$\varepsilon_{\text{dir}} = \frac{|V - V^{\text{ref}}|}{|V^{\text{ref}}|}.$$

The corresponding two-sided $(1 - \alpha)$ confidence interval is reported as

$$\left[V - z_{1-\alpha/2}\,\text{SE},\ V + z_{1-\alpha/2}\,\text{SE}\right],$$

with $\alpha = 0.05$. Table 2 gives the benchmark prices, standard errors, and confidence intervals used throughout the study.

Table 2: Direct benchmark reference values used in the numerical comparisons.

K	Direct price	SE	Confidence interval
70	2.699339	0.007474	[2.684690, 2.713988]
80	4.596899	0.009874	[4.577546, 4.616252]
90	7.402999	0.012459	[7.378579, 7.427419]
100	11.356823	0.015058	[11.327309, 11.386337]
110	16.645784	0.017243	[16.611988, 16.679580]

8.2 Fixed path budget, varying number of steps

This experiment investigates how the number of steps affects the direct Bermudan price when both methods are run with the same path budget. The path budgets 20,000 and 60,000 are intentionally moderate, so that Monte Carlo variation is still visible and the effect of the conditional PDE step can be assessed away from the benchmark regime.

Matched 20,000**-path comparison.** The first comparison uses 20,000 direct paths for both methods and step counts 24, 48, 72, and 96. Figure 1 reports the relative errors across the five strikes. The results show that the Hybrid LSMC–PDE method gives smaller relative error at all four step levels for $K = 80$, $K = 90$, $K = 100$, and $K = 110$, and at three of the four step levels for $K = 70$. The difference between the two methods is therefore clearest at this lower path budget, especially for the out-of-the-money and at-the-money contracts.

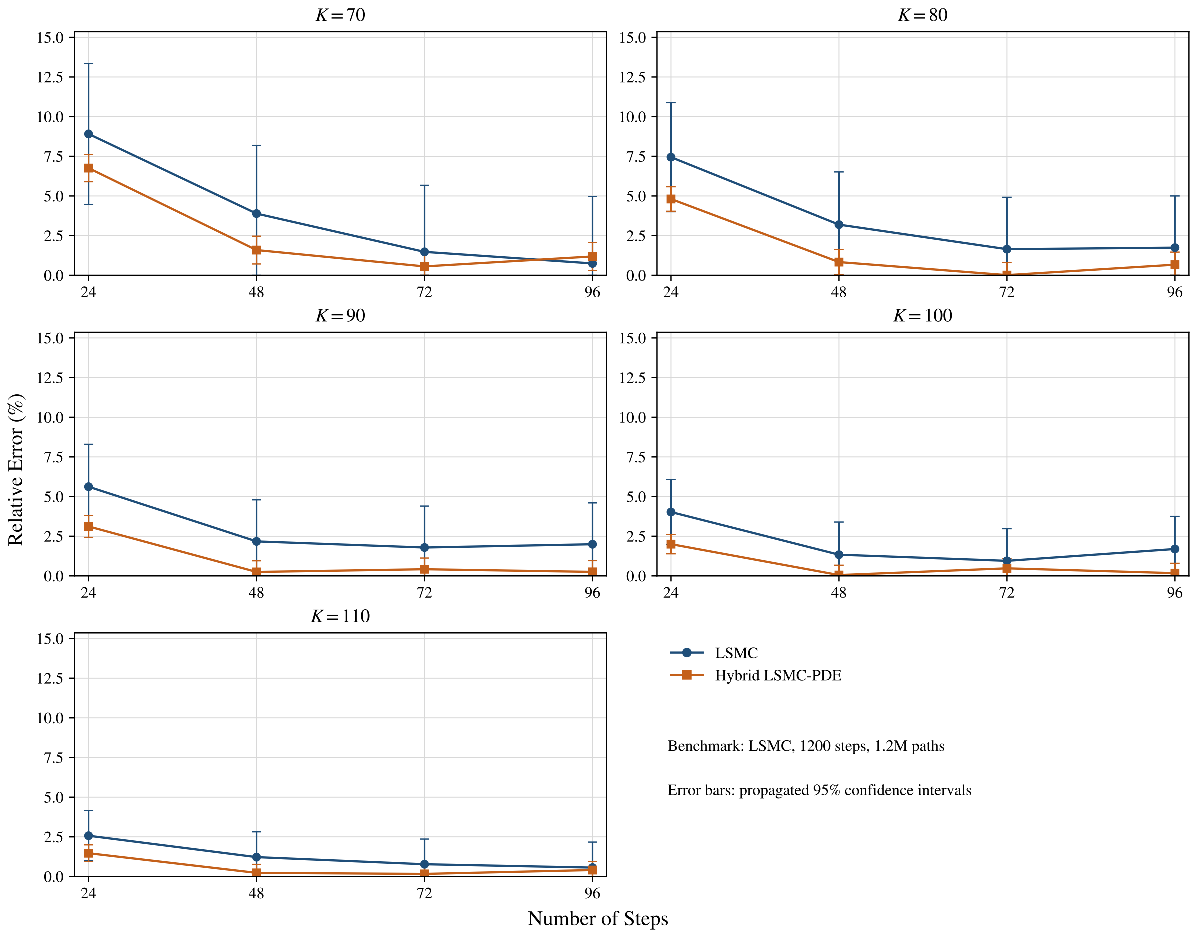

Fig. 1 Relative errors for the matched 20,000-path step-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Table 3 reports representative results at 72 steps. The improvement is visible for all strikes. The LSMC relative errors lie between 0.773% and 1.788%, while the Hybrid LSMC–PDE relative errors lie between 0.012% and 0.560%. The standard errors are also smaller under the hybrid method, and the confidence intervals are correspondingly narrower. This shows that, under the same path budget, the hybrid direct prices are both closer to the benchmark and less dispersed.

Table 3: Representative direct metrics at 72 steps under the matched 20,000-path comparison.

	LSMC			Hybrid LSMC–PDE		
K	Dir. err.	SE	Confidence interval	Dir. err.	SE	Confidence interval
70	1.476%	0.057802	[2.625881, 2.852465]	0.560%	0.012206	[2.690521, 2.738368]
80	1.645%	0.076738	[4.522131, 4.822943]	0.012%	0.018645	[4.560886, 4.633975]
90	1.788%	0.098666	[7.342003, 7.728773]	0.418%	0.026674	[7.319752, 7.424315]
100	0.948%	0.117430	[11.234369, 11.694695]	0.475%	0.035909	[11.232454, 11.373219]
110	0.773%	0.134824	[16.510179, 17.038689]	0.163%	0.045393	[16.529605, 16.707546]

Matched 60,000-path comparison. The second comparison increases the common path budget to 60,000 while keeping the same step grid. Figure 2 shows that the Hybrid LSMC–PDE method again performs best at the intermediate step levels. The difference between the two methods is clearest at 48 and 72 steps across the full strike set, while the gap becomes smaller at 24 and 96 steps.

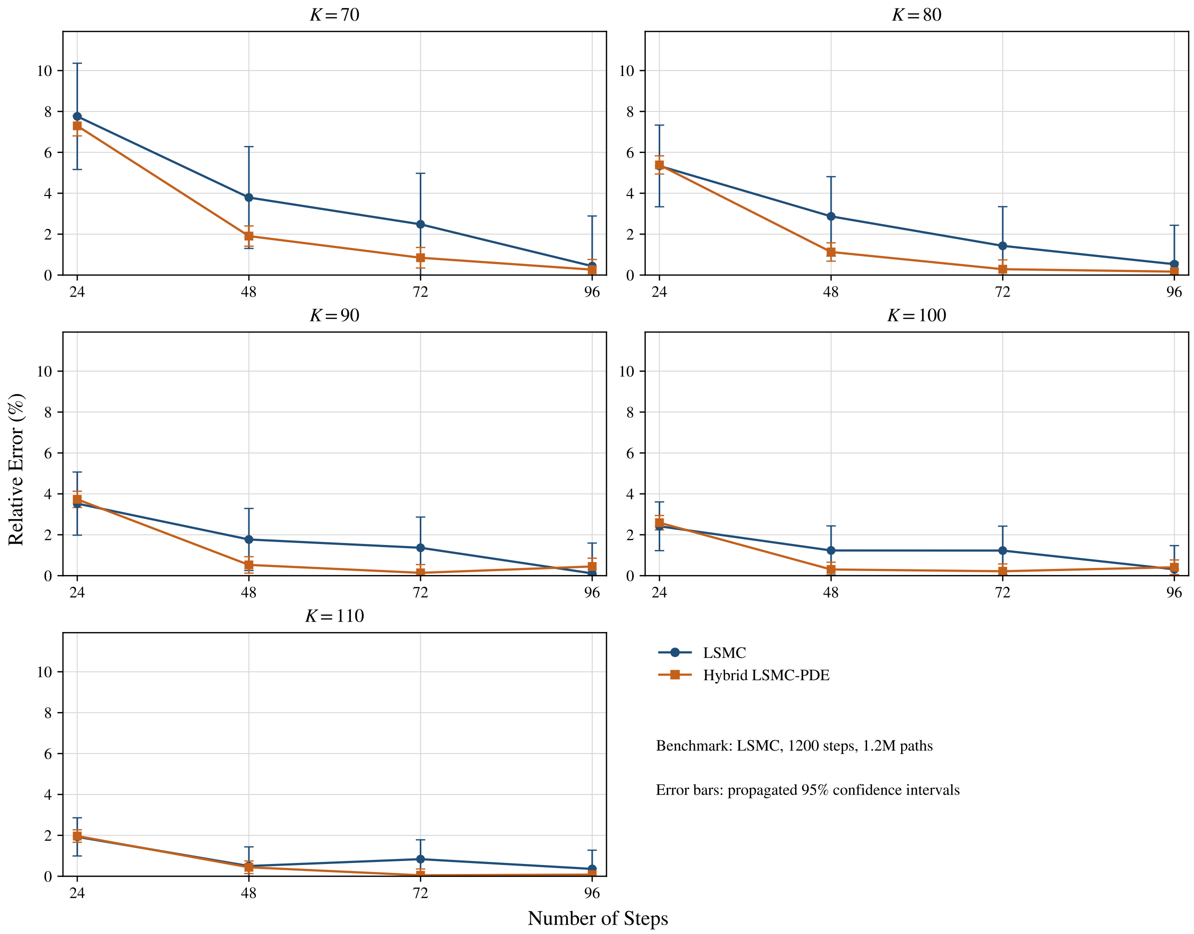

Fig. 2 Relative errors for the matched 60,000-path step-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Table 4 reports representative results at 72 steps. At this resolution, the LSMC relative errors range from 0.837% to 2.482%, while the Hybrid LSMC–PDE relative errors range from 0.048% to 0.847%. The standard errors and confidence intervals are again smaller under the hybrid method. Hence, the improvement in accuracy is accompanied by lower sampling variability.

Table 4: Representative direct metrics at 72 steps under the matched 60,000-path comparison.

	LSMC			Hybrid LSMC–PDE		
K	Dir. err.	SE	Confidence interval	Dir. err.	SE	Confidence interval
70	2.482%	0.034333	[2.699045, 2.833631]	0.847%	0.006908	[2.708657, 2.735737]
80	1.429%	0.044924	[4.574527, 4.750629]	0.287%	0.010629	[4.589249, 4.630917]
90	1.363%	0.056750	[7.392681, 7.615141]	0.139%	0.015294	[7.362757, 7.422707]
100	1.227%	0.069157	[11.360645, 11.631741]	0.216%	0.020668	[11.291787, 11.372805]
110	0.837%	0.080321	[16.627687, 16.942545]	0.048%	0.026208	[16.602442, 16.705177]

The matched-path experiments lead to the same overall conclusion. The advantage of the Hybrid LSMC–PDE method is most visible at intermediate step levels. It is also more pronounced when the path budget is reduced from 60,000 to 20,000, so the difference between the two estimators is clearest in the moderate-sample regime. A few reversals remain at 24 and 96 steps, mainly in the deepest out-of-the-money case.

8.3 Fixed number of steps, varying path counts

This experiment complements the step-sweep study by fixing the number of steps and varying the direct path budget. The reported path counts are 250, 1000, 5000, 10000, 20000, 40000, and 60000. This range covers the low-sample regime, where regression noise is still substantial, together with the moderate- and larger-sample regimes in which the Monte Carlo decay becomes visible. Two moderate step levels, 48 and 60, are examined.

Fixed 48**-step comparison.** The first path-sweep experiment fixes 48 steps and varies the direct path count over 250, 1000, 5000, 10000, 20000, 40000, and 60000. Figure 3 shows the corresponding relative-error curves. For both methods, the relative-error envelope decreases as the number of paths increases, although finite-sample fluctuations are still visible at the smallest path counts. The Hybrid LSMC–PDE method has smaller relative error for all reported path counts and for all five strikes. The improvement is already visible at 250 paths and remains uniform across the full reported path grid.

Table 5 reports representative results at 20,000 paths. The LSMC relative errors range from 1.221% to 3.890%, whereas the Hybrid LSMC–PDE relative errors range from 0.052% to 1.589%. The hybrid estimator is already below 1% for four of the five strikes. The standard errors and confidence intervals are also uniformly smaller under the hybrid method, which indicates lower sampling variability at the same path budget.

Table 5: Representative direct metrics at 20,000 paths under the fixed 48-step path sweep.

	LSMC			Hybrid LSMC–PDE		
K	Dir. err.	SE	Confidence interval	Dir. err.	SE	Confidence interval
70	3.890%	0.059206	[2.688290, 2.920378]	1.589%	0.012069	[2.718589, 2.765901]
80	3.193%	0.077914	[4.590982, 4.896404]	0.832%	0.018497	[4.598892, 4.671399]
90	2.173%	0.098998	[7.369862, 7.757934]	0.248%	0.026580	[7.369274, 7.473469]
100	1.336%	0.119290	[11.274695, 11.742311]	0.052%	0.035911	[11.292378, 11.433149]
110	1.221%	0.135565	[16.583297, 17.114711]	0.228%	0.045516	[16.594577, 16.773000]

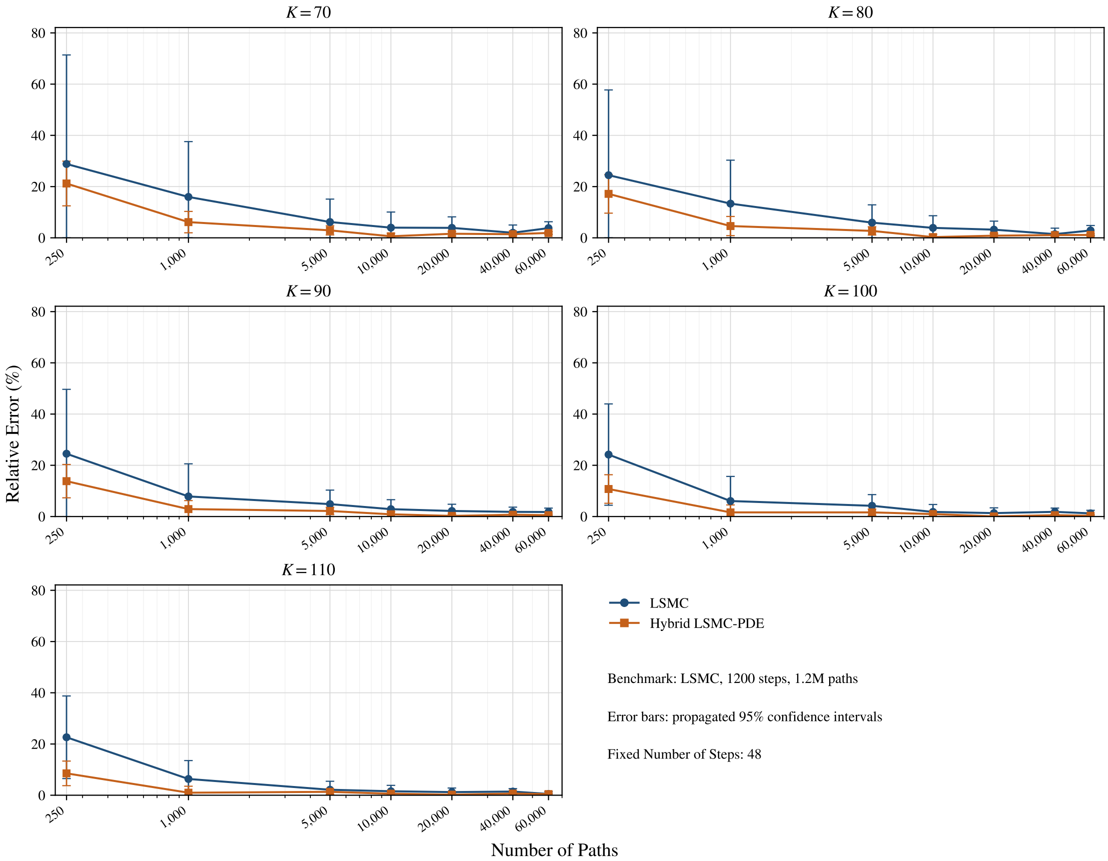

Fig. 3 Relative errors for the fixed 48-step path-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Fixed 60-step comparison. The second path-sweep experiment keeps the same path grid and increases the discretization to 60 steps. Figure 4 shows that the hybrid advantage remains visible across the reported path grid. The Hybrid LSMC–PDE estimator has smaller relative error for all reported path counts and for all five strikes. The separation between the two methods is most visible in the low- and moderate-path regimes, while the gap narrows as the path budget increases.

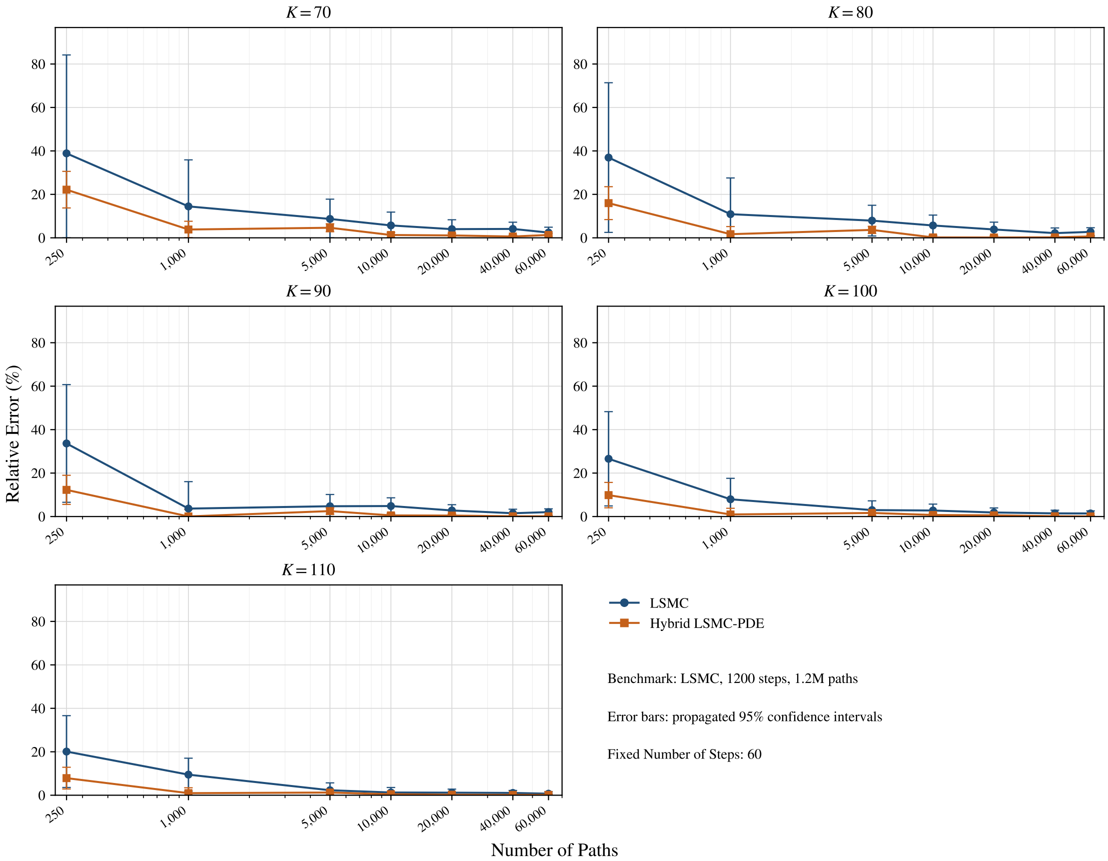

Fig. 4 Relative errors for the fixed 60-step path-sweep experiment. Error bars are induced by the 95% confidence intervals of the direct estimators.

Table 6 reports representative results at 20,000 paths. The same pattern is observed. The LSMC relative errors lie between 1.177% and 3.997%, while the Hybrid LSMC–PDE relative errors lie between 0.156% and 1.100%. The hybrid estimator is below 1% for $K = 80$, $K = 90$, $K = 100$, and $K = 110$. The standard errors and confidence intervals are again smaller under the hybrid method. The ranking between 48 and 60 steps is not perfectly uniform across moneyness, but the main comparison between methods remains unchanged.

Table 6: Representative direct metrics at 20,000 paths under the fixed 60-step path sweep.

	LSMC			Hybrid LSMC–PDE		
K	Dir. err.	SE	Confidence interval	Dir. err.	SE	Confidence interval
70	3.997%	0.059501	[2.690621, 2.923865]	1.100%	0.012465	[2.704611, 2.753474]
80	3.847%	0.078962	[4.618994, 4.928526]	0.156%	0.019048	[4.566717, 4.641386]
90	2.791%	0.100311	[7.412982, 7.806202]	0.450%	0.027259	[7.316246, 7.423100]
100	1.864%	0.120950	[11.331454, 11.805578]	0.558%	0.036673	[11.221571, 11.365327]
110	1.177%	0.138568	[16.570147, 17.113333]	0.239%	0.046322	[16.515281, 16.696863]

Both study families point in the same direction. In most cases, the Hybrid LSMC–PDE method produces direct prices that are closer to the benchmark and accompanied by smaller standard errors and narrower confidence intervals. The improvement is most visible at moderate path budgets and at intermediate step levels. The few reversals that remain are confined to the fixed-path step-sweep study, mainly in the deepest out-of-the-money case. Overall, the numerical results show that the conditional PDE step improves both accuracy and sampling stability for the direct Bermudan estimator under the calibrated gDMR specification.

9 Conclusion

Acknowledgments

The authors thank Anatoliy Malyarenko for helpful discussions.

Declarations

Funding

Not applicable.

Conflicts of interest/Competing interests

The authors declare that they have no competing interests.

Availability of data and materials

Not applicable.

Code availability

/

Authors' contributions

To be written. All authors read and approved the final manuscript.

References

[1] Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973. doi: 10.1086/260062.

[2] K. C. Chan, G. Andrew Karolyi, Francis A. Longstaff, and Anthony B. Sanders. An empirical comparison of alternative models of the short-term interest rate. *The Journal of Finance*, 47(3):1209–1227, 1992. doi: 10.1111/j.1540-6261.1992.tb04011.x.

[3] Peter Christoffersen, Steven Heston, and Kris Jacobs. The shape and term structure of the index option smirk: Why multifactor stochastic volatility models work so well. *Management Science*, 55(12):1914–1932, 2009. doi: 10.1287/mnsc.1090.1065.

[4] Emmanuelle Clément, Damien Lamberton, and Philip Protter. An analysis of a least squares regression method for American option pricing. *Finance and Stochastics*, 6(4):449–471, 2002. doi: 10.1007/s007800200071.

[5] John C. Cox. Notes on option pricing I: Constant elasticity of variance diffusions, 1975. URL `https://www.jstor.org/stable/2327490`. Working paper, Stanford University.

[6] John C. Cox, Stephen A. Ross, and Mark Rubinstein. Option pricing: A simplified approach. *Journal of Financial Economics*, 7(3):229–263, 1979. doi: 10.1016/0304-405X(79)90015-1.

[7] John C. Cox, Jonathan E. Ingersoll, and Stephen A. Ross. A theory of the term structure of interest rates. *Econometrica*, 53(2):385–407, 1985. doi: 10.2307/1911242.

[8] Andrei Cozma and Christoph Reisinger. A mixed monte carlo and partial differential equation variance reduction method for foreign exchange options under the Heston–Cox–Ingersoll–Ross model. *The Journal of Computational Finance*, 20(3):109–149, 2017. doi: 10.21314/JCF.2016.318.

[9] David Farahany. *Mixing Monte Carlo and Partial Differential Equation Methods for Multi-dimensional Optimal Stopping Problems Under Stochastic Volatility*. PhD thesis, University of Toronto, 2019. URL `https://www.cs.toronto.edu/pub/reports/na/Farahany_David_PhD_Thesis.pdf`.

[10] David Farahany, K. R. Jackson, and Sebastian Jaimungal. Mixing monte carlo and PDE methods to price bermudan options under stochastic volatility. *SIAM Journal on Financial Mathematics*, 11(1):201–239, 2020. doi: 10.1137/19M1249035.

[11] Jim Gatheral. *The Volatility Surface: A Practitioner's Guide*. Wiley, 2006.

[12] Jim Gatheral. *The Volatility Surface: A Practitioner's Guide*. John Wiley & Sons, 2006. ISBN 9780471792512. URL https://www.wiley.com/en-us/The%2BVolatility%2BSurface%3A%2BA%2BPractitioner%27s%2BGuide-p-9780471792512.

[13] Paul Glasserman. *Monte Carlo Methods in Financial Engineering*, volume 53 of *Stochastic Modelling and Applied Probability*. Springer, 2004. doi: 10.1007/978-0-387-21617-1.

[14] Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *The Review of Financial Studies*, 6(2):327–343, 1993. doi: 10.1093/rfs/6.2.327.

[15] Patrick Jaillet, Damien Lamberton, and Bernard Lapeyre. Variational inequalities and the pricing of american options. *Acta Applicandae Mathematicae*, 21:263–289, 1990. doi: 10.1007/BF00047211.

[16] Ioannis Karatzas and Steven E. Shreve. *Brownian Motion and Stochastic Calculus*, volume 113 of *Graduate Texts in Mathematics*. Springer, 2 edition, 1991. doi: 10.1007/978-1-4612-0949-2.

[17] Thomas Lipp, Grégoire Loeper, and Olivier Pironneau. Mixing Monte-Carlo and partial differential equations for pricing options. *Chinese Annals of Mathematics, Series B*, 34(2):255–276, 2013. doi: 10.1007/s11401-013-0763-2.

[18] Grégoire Loeper and Olivier Pironneau. A mixed PDE/Monte-Carlo method for stochastic volatility models. *Comptes Rendus Mathematique*, 347(9–10):559–563, 2009. doi: 10.1016/j.crma.2009.02.021.

[19] Francis A. Longstaff and Eduardo S. Schwartz. Valuing American options by simulation: A simple least-squares approach. *The Review of Financial Studies*, 14(1):113–147, 2001. doi: 10.1093/rfs/14.1.113.

[20] Robert C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183, 1973. doi: 10.2307/3003143.

[21] Robert C. Merton, Michael J. Brennan, and Eduardo S. Schwartz. The valuation of american put options. *The Journal of Finance*, 32(2):449–462, 1977. doi: 10.1111/j.1540-6261.1977.tb03284.x.

[22] Yuliya Mishura, Andrey Pilipenko, and Kostiantyn Ralchenko. Gatheral double stochastic volatility model with Skorokhod reflection, 2025.

[23] John N. Tsitsiklis and Benjamin Van Roy. Regression methods for pricing complex American-style options. *IEEE Transactions on Neural Networks*, 12(4):694–703, 2001. doi: 10.1109/72.935083.

[24] Toshio Yamada and Shinzo Watanabe. On the uniqueness of solutions of stochastic differential equations. *Journal of Mathematics of Kyoto University*, 11(1):155–167, 1971. doi: 10.1215/kjm/1250523691.